Rules of Derivatives

1. Constant Rule: The derivative of a constant k is zero, $\frac{d}{dx}k = 0$.

2. Power Rule: The derivative of kx^n, where k and n are constants, is $\frac{d}{dx}(kx^n) = knx^{n-1}$.

3. Sum Rule: The derivative of a sum is the sum of the derivatives, $\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$.

4. Product Rule: The derivative of a product is $\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$.

5. Quotient Rule: The derivative of a quotient is $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$, $g(x) \neq 0$.

6. Chain Rule: The derivative of a function nested inside another function is

$$\frac{d}{dx}[f(g(x))] = f'(g(x))g'(x).$$

7. Natural Log Rule: The derivative of the natural log of a function is

$$\frac{d}{dx}\ln(f(x)) = \frac{f'(x)}{f(x)} \quad \text{where } f(x) > 0.$$

8. Exponent Rule: The derivative of an exponential function with base e is

$$\frac{d}{dx}e^{f(x)} = f'(x)e^{f(x)}.$$

9. Total Derivative: For a function of two variables, $F(x, y)$, the total derivative is

$$dF(x, y) = F_x\,dx + F_y\,dy \quad \text{or equivalently,} \quad dF(x, y) = \frac{\partial F}{\partial x}\,dx + \frac{\partial F}{\partial y}\,dy.$$

The symbol ∂ denotes a partial derivative. Partial derivatives measure how a function changes with respect to one variable while holding the other variables constant.

10. Implicit Function Theorem:

Suppose $F(x, y)$ is held constant along a level set, so $F(x, y) = c$. Because F is constant along this curve, its total derivative is 0.

$$dF(x, y) = F_x\,dx + F_y\,dy = 0.$$

With some algebra,

$$F_y\,dy = -F_x\,dx \implies \frac{dy}{dx} = -\frac{F_x}{F_y} \quad \text{or equivalently,} \quad \frac{dy}{dx} = -\frac{\frac{\partial F}{\partial x}}{\frac{\partial F}{\partial y}}$$